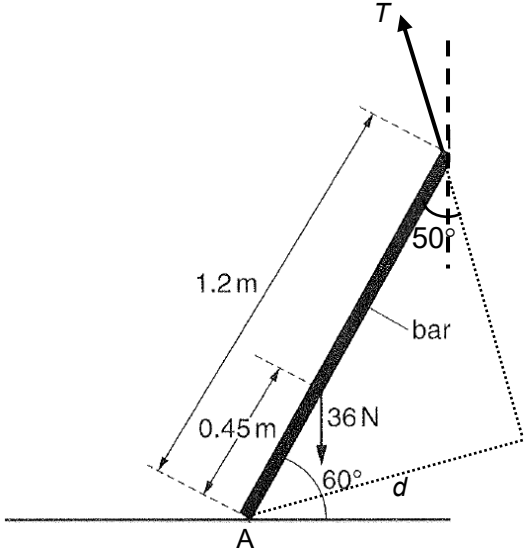


No.	Solution	Remarks
1(a)(i)	$B = \frac{\phi}{A} = \frac{50 \times 10^{-6}}{4.00 \times 10^{-4}} = 0.125 \text{ T}$ $\frac{\Delta B}{B} = \frac{\Delta \phi}{\phi} + \frac{\Delta A}{A} = \frac{5}{50} + \frac{0.01}{4.00} = 0.1025$ $\Delta B = (0.125 \text{ T})(0.1025)$ $= 0.01 \text{ T (1 s.f.)}$ $\rightarrow B = (0.13 \pm 0.01) \text{ T}$	[1] value of $\frac{\Delta B}{B}$ [1] value of ΔB to 1 s.f. [1] answer
1(a)(ii)	The percentage (or fractional) uncertainty of ϕ is more significant than that of A .	[1]
1(b)(i)	 The perpendicular distance from A to line of action of T is given by $d = 1.2 \sin 50^\circ$ Taking moments about A, $(36)(0.45 \cos 60^\circ) = T (1.2 \sin 50^\circ)$ $T = \frac{(36)(0.45 \cos 60^\circ)}{1.2 \sin 50^\circ}$ $= 8.81$ $= 8.8 \text{ N (shown)}$ OR Anti-clockwise moment $= T_y (1.2 \cos 60^\circ) + T_x (1.2 \sin 60^\circ)$ $= (T \cos 20^\circ)(1.2 \cos 60^\circ) + (T \sin 20^\circ)(1.2 \sin 60^\circ)$	[1] moment equation [1] numerical answer before 8.8 N

No.	Solution	Remarks
1(b)(ii)	Let F_x and F_y be the horizontal and vertical components of F respectively. Consider vertical forces: $36 \text{ N} = T \cos 20^\circ + F_y$ $\rightarrow F_y = 36 - 8.81 \cos 20^\circ = 27.72 \text{ N}$ Consider horizontal forces: $F_x = T_x = T \sin 20^\circ$ $= 8.81 \sin 20^\circ$ $= 3.01 \text{ N}$ $F = \sqrt{F_x^2 + F_y^2} = \sqrt{3.01^2 + 27.72^2} = 27.9 \text{ N}$	[1] value of F_y [1] value of F_x [1] answer
2(a)	The phase difference is due to the extra distance x travelled by	